

ActInf Textbook Group ~ Cohort 4 ~ Meeting 16 (Chapter 7 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 16 (Chapter 7 part 1) | 57:42

all right greetings everyone it is

October 10th 1010

23 and we're in our discussion of

chapter 7 so last week Ali and I

recorded like the short video overviews

of chapters nine and 10 just because

there was um only a few of us during our

cohort for slot and we just wanted to

complete those um overviews so now we're

in the chapter

7 um discussion so we'll jump over to

the chapter 7

page um and before we look at like any

of the specific questions like does

anybody just want to

share

um what did they think about chapter 7

what what was there impression or

or how would they summarize it to to

somebody or what was just exciting to

them about chapter

7 well can I start

um

so

please

hello Daniel hi yeah go forward go

forward yeah so so my um impression of

chapter 7 is about

uh uh it's many about about the uh

discrete uh

State and discrete outcome model of the

active

inference

and I found it is particularly uh useful

to have a detail understanding

of one of the example which is this

T's uh

example and it is kind of the first uh
active inference model that I can get my
hand on and try it out and change those
uh those parameters or the the matrixes
used in the active inference model and
see how

um how the uh active inference uh
actually

works awesome I think that's kind of the
point

um anyone else want to give a thought on
seven um yeah I would just second

everything that uh Lan said just now um

I think it's a great way to actually see
maybe how some of the the ideas that
we've been reading about prior in the
textbook actually start to hang together
and play out in an example I think that
the the use of discret variables is
highly like kind of interpretable I

think this is a it comes a little bit
more naturally to me to read this
chapter

versus um you know chapter 8 uh with
continuous variables because we get to
see like your you know more precise
specific things that happen discret
choices and and things that occur and
then also just some kind of interesting
observations that come out of that with
the different um kind of components of
the model like uh you know as you get
further in the chapter there's a
discussion of like here are the
interesting things that happen whenever
you don't include preferences or have

highly uh uniform preferences like
there's just pure novelty seeking on the
part of the agent who like um like in
the team's example they just can't sit
still they never come back to the same
spot they just it's pure just seeking
epistemic value it seems because there's
nothing pragmatic to do because there's
nothing in particular that they're going
for because it's not been hardcoded into
the initialized um preferences so so
yeah just um giving us that that really
big model for active inference here in
the discret form in addition seeing the
interesting dynamics that can kind of
play out emerging properties and so
on

awesome anyone else want to make a
general comment on
seven I would I would just Echo that
given that I've come from a psychology
background and so seeing the exploration
exploitation Paradigm just being laid
out in that way and they make the
comment that you know they believe
they've

uh come to understand how these two go
hand in hand instead of being
competitive I thought that was also very
very

intuitive cool yes yeah the continuous
time setting is kind of like having your
hand on the rudder in a
boat and uh so you're making continuous
actions and there's a continuous flow in
these paths
and the discrete time is like chess or
Checkers

like time

clicks discreetly and the actions are
discreet the states are discreet and so
that can make it kind of clearer a
little bit more cut and dry

um cool does anybody want to jump to a
specific section of seven or we can look
at the questions and just like look at
at the upvoted ones or if anybody
submitted any and just like look at
what's interesting in
seven or or somebody can propose a
section that that they want to look
at I'm yeah I'm mostly interest see in
chapter uh

uh

7.3 uh decision making and planning as a
reference uh I wonder if you know if
someone else has preference to talk
about the sure well

let's yeah let's get to 7.3 basically

7.1 is just one paragraph we can see

here 7.2 is just a passive inference

model this is just the downstairs

section of the discrete time model model

we have hidden States changing through

time being related to observations at

each time point and a prior so it's just

the downstairs of figure

4.3 and it's kind of worked out in

multiple stages with this music

listening example so then we get to

7.3 now we take that downstairs the part

that looks like an

e um and we add in the action so

so this is where we actually get active

inference and then what makes it active

inference under the free energy

principle is that the unified imperative for Action selection is going to be expected free energy so right because there are other models that have um sense making or signal processing on the inbound and then decision- making or action selection or control theory on the outbound

like that is in a loose sense active inference but when we're talking about active inference we're talking about active inference and the free energy principle which is what we get when we bring in the variational and the expected free energy okay so what what aspect of 7.3 do you want to explore well first about this figure that you're are showing here um I I have a question about uh the fourth line from the

bottom uh you you see third while c and e appear in the panel on the right do you see that sentence where is the e i I couldn't find the

E yes so e um is the Habit e represents the prior Distribution on policies so let's just say that we have four affordances up down left right the E Vector is like 0.25 0.25 0.25 0.25 so in that

situation e conveys through its inclusion which actions are possible and okay by the value in E that is the prior on policy so

if they were all 0.25s and there was four um

affordances then that would be like
saying those options um in the absence
of being sharpened by expected free
energy are equally likely to be
taken and
um we can look at um the the nervous
system diagram in
figure 5.4 so here's e so here on the
indirect
pathway observations are coming in
habits are being
triggered on the direct pathway we have
expected free energy
sharpening our habit
distribution that doesn't just mean yeah
it doesn't just mean exaggerating it it
could upweight something like let's just
say there's a rare policy um like but a
given situation it has high epistemic
Andor pragmatic value well then you'd
hope that even though your prior on um
doing it is very low like what's the
prior on taking off your seat belt well
on a 10hour car ride it's low but then
when the car is stopped and you want to
leave that's an amazingly Salient thing
to do so e basically hosts what can be
done and in the absence of any
decision-making process the distribution
of actions to be taken and then G
sharpens
that to to to to update the policy
posterior and then the policy posterior
is where action is sampled
from I see I see from from the figure
7.3 it seems that c and e plays the same
role as f in the distribution of poliy
right they kind

of Absol R um absolutely like um free
energy expected free energy here G can
be understood as a statistical
distribution about
policy conditioned upon
CN we we just discussed why e matters
for the calculation of policy posterior
C matters because of equation
2.6 so C matters for G calculation
because C is going to decide what is of
pragmatic
value so um in that representation that
they have in
7.3 basically just
saying um policy is being computed
conditioned
upon
preference and on
habit okay so C is preference e is habit
so they they encode different uh kind of
preferences of the policies
yeah okay thank you denner that's very
very good cool yeah I mean anyone else
have like a thought on 7.3 or or a
question on 7.3 or just how how they see
it or how they interpret
it so I I have further question so again
on this figure 7.3 so we see the policy
 P_i affects the um the the um probity uh
State probity transition Matrix
B uh I wonder uh if it's possible to
model the uh policy also affect the
Matrix a meaning that it it it not only
affects the perp transition mes Matrix
but also um affects the transformation
from the uh state to the
outcome great great question all all
kind of mention two ways that you could

think about that so as you mentioned policy intervenes in B how things change through time like when you turn on the air conditioner it doesn't like reach into the room and just like modify the thermometer it modifies how the temperature changes through time so it modifies the Dynamics of the Hidden State the temperature and then that results in a thermometer changing so first answer to the question is we don't need policy to change a if we want if we prefer it to be cooler and we take the policy to turn on the air conditioner then at in the next time step the temperature of the room is going to drop let's say one policy is you know turn on the air conditioner temperature drops by one turn off it it goes up by one so you turn it on the temperature drops in the room and then the the thermometer is cooler so one answer is you don't really want to touch a because the mapping between hidden States and observations isn't really something you can act on directly and so we're intervening in the real underlying hidden causal process and then that emits OB the first and the second answer is you could develop an active inference model where a metacognitive agent selects which a matrix to use so um maybe you decide whether to put on sunglasses or not and then when you have sunglasses

on colors are mapped to your vision this way and when you do something different they're mapped a different way so that would be taking a policy choice on which a to

use okay

but in most settings the relationship between hidden States and observations is um it's orthogonal to or you could say it's factorized away from how hidden States change through time keeping this model sparse in the sense that keeping as few edges as possible keeps the computational cost low and keeps the um kind of parsimony High let's see thanks than that's very good so

um yeah so so it really depends on how the

states and the outcomes are are defined

um maybe in my case we can Define it such way that that high the policy affects only the states transition perity not affecting the

um the trans Nation from the state to outcomes yeah in in almost all cases you will have policy only

intervening in B so making a generative model I mean this chapter six making a generative model is if you can describe what all of these are in your system of interest you literally have the generative model and how do you categorize different things that you have or want to include into this well observations are literally the data if it's in a spreadsheet it's literally an

observation then right policies are actions so if something is something that an agent can do and then everything else is a hidden

State now if you're in a fully observable situation like a chessboard then it can seem like there's a

redundancy like the observation is the location of the piece and the hidden state is the true location of the piece so in that case it's not really a partially observable Markov decision process pomdp it's just an

mdp because it's a fully observable decision process but a allows us to bring in the partial observability but if you're in a fully observable

setting then you can almost think about direct

intervention however even in that case then the A matrix is just an identity Matrix like it Maps perfectly between hidden States you know you have an accurate thermometer 37 just maps to 37 then A is an identity Matrix and it has basically no computational

cost and then your model is more flexible in case you ever did have a noisy thermometer or a missing measurement if you had a

pmdp then it's fine but if you were to only have the mdp then you can't deal with the noisy or a partially observable setting yes thank you thank you Dan

that's very detailed
exper

yeah I'll hold on uh hold my questions
for okay other participants to ask yeah
anyone want to just give a a thought or
or a question on this
section um yeah I guess just since it's
it's been a bit since I actually read
like really did a deep dive on this
particular chapter um but um I just I'm
trying to make sure I grasp like
whenever they're describing the the
teamas shortly after um yeah I guess
we're still in 7.3 they just start to
describe like the a matrices is like in
this case they become
tensors um like with the teamat you have
there are outcomes and then there are
locations and then there are contexts so
I guess the you know the the mouse uh
searching for a reward is looking for I
guess the outcome is like whether the
the space they land on is a reward or
not um and um the location is just you
know them navigating the maze itself and
the locations of the Maze and then also
you have the the context I might have I
might be mixing those up a bit but it's
just
just trying to make sure I can grasp
like we're looking at potentially 3D 4D
and so on um like kind of tensor um you
know quantities here that are based on
all of the kind of Dimensions to what
what the the mouse is doing rather than
just uh yeah looking at as like a 2d
thing as As Nice and simple as that
would
be yes um
ba basically yes so a I mean you could

think of it as okay well pull back the context is whether you're in um world State a or World State B so the two contexts are whether the the food is on the left or the right um in in in in in this setting you just you did mention that um okay so now a Maps between observations and hidden stat States there's going to be two types of hidden States and so each of them projects out to a given

observation

A1 is about location A2 is about context I don't know if you would

properly call this a

tensor but certainly this is two

matrices um but it it's not but they're

different shapes this is a 5x5 Matrix

and this is a a a 3x4 matrix when we get

to B it will be a true tensor in the

sense that every slice of B is going to

be a matrix with the same

dimensionality but here it's kind of

like two parallel channels like this

would be analogous to having two sensory

modalities so if if we had had vision

and hearing those a

matrices um could map to the same hidden

State like maybe you know is there

lightning um or is is there a

thunderstorm and you might have a

mapping between hearing and is there a

thunderstorm um and vision and is there

a thunderstorm or you can have those two

observation

modalities maap to two different hidden

State modalities like in this case I

just want to make one more point before

we continue um forward in the teas
because um it was brought up about the
explore

exploit the most probable policies are
those that lead to the lowest expected
free energy again this is one of the key
distinguishing features if not the key
distinguishing feature between active
inference guided policy selection and
reinforcement learning or reward
maximization there is no proposal of
a reward or utility function that gets
optimized there's just the definition of
the generative model and then the
generative model does the likeliest
thing given its setup again that doesn't
mean the animal lives it doesn't mean
that XYZ

it just means that given how the
generative model is set up it does the
likeliest thing and then given
empirical data chapter nine we set up or
parameterize the generative model to be
the likeliest

one so Daniel can yeah go Ahad sorry
yeah I I do have a question regarding
744 where you're right there yeah here
uh so I'm I be confused what p and Q
means um do we always use P represents
the uh prior or the true distribution
and Q as a approximate distribution Q is
is is this Q the one we actually want uh
the
decision that we uh we intend to um to
change our belief by
updating uh
approximation to the uh real real
probability so p is the real probability

exactly Q is the real mapping in this case um so it's it's a natural log so this is kind of like taking a surprisal okay it's a it's of observations conditioned upon preferences so this is really what What You observe in relationship to what you prefer you and the reason why it it has the p is because we're talking about like the the actual observations coming in and Q is the variational distribution that we control okay okay yeah now it's clear so actually we can change Q to minimize this GQ GP this object function here right yeah um yeah like Q are about our beliefs whereas we don't really get to control in the same way like we Q of observations conditioned upon policy so what do I believe the sequence of future observations will be i lda conditioned upon what I do that's all in the mind that is different than what is the sequence of observations given my preferences you can change your preferences but that doesn't change observations given your preferences and similarly you could change your belief about hidden States but you can't in the same way control the mapping between observations and hidden States in actuality yeah and this is like a restatement in equation 7.4 of equation 2.6 expected free energy so we have same epic and pragmatic value okay so let's go through the teamas and just see how this is going to play out okay so the

two factors so factor is being used to describe a different kind of hidden State um so again there's two factors here there's the context factor and the location factor

one of those is a controllable Factor location

whereas context is not a controllable Factor if it were there would be another policy

type change the food to the left to the right but we're not in that setting so this example brings up the distinction between controllable and uncontrollable factors

observations they are the fundamental observables that's in the spreadsheet then everything that's not an observation is a hidden State Factor State factor and then some State factors are controllable those are the ones that have policies intervening in their B matrices other state factors are uncontrollable it's just a latent cause of the world that's proposed within a cognitive model but with respect to that uncontrollable State Factor that's like listening to music the first example that's like a passive inference on that state

Factor okay so the controllable state are the uh are the locations of the mouse right right because it can move uncontrollable states are the locations of this

yeah because the mouse doesn't get to decide where the food is right yeah okay yeah um so that then

that's where agency and self-efficacy
can come into play
is to what extent do actions bear
upon
um Can can the agent take actions and
make a difference that makes a
difference take an action that changes
some state in the world controllable
factor
and how are and then this is all about
how they evaluate that it could be
chosen because it has high pragmatic
value because doing that kind of
exerting that kind of control brings
observations closer to alignment with
preferences pragmatic value
or it might lead to new information
about the world epistemic value and
that's the entropy the
expected
difference of the
entropies
of observations conditioned upon hidden
States in comparison with
observations conditioned upon
policy so this is basically like the
entropy of the a
matrix so that's that's the mapping
between observations in Hidden States
here we're asking how much is it going
to matter differently if we condition
upon policy so let's just say that the a
matrix has a a given entropy the a
matrix has a given uncertainty
associated with it the least uncertain a
matrix is like the identity Matrix just
one on the diagonal and all zeros
elsewhere you have perfect knowledge it

becomes a fully observable setting the most entropy a matrix is just All Uniform basically the measurements don't even matter because all the measurements are equally likely to map to any hidden state so that's on the left side and now the question is conditioned upon action is there going to be a difference in the sharpness of that mapping well if it's a policy that doesn't lead to sharpening of the a it doesn't have epistemic value in comparison if there's a policy such that you believe that taking that policy is going to sharpen your understanding of outcomes then this is a non-zero value so that's the epistemic value yes Daniel the um however you said that you know from the model that a matrix is fixed right it's not affected by you know all the entries in a matrix are not affected by the P_{CY} P_i because P_{CY} P_i affects only the state transition probability Matrix

B and what do you mean by sharpen a using policy

P_i good question so there's there's a few um a few pieces here first off there's a matrix learning that is not happening in this example so okay but but but but it is it is a cognitive phenomena that the a matrix could be updated but it's not being there's no learning here um now let's just say that there's um we have we have a a two observ we have observations and and hidden States one of the

observations doesn't lead to gain
information about hidden

States like it Maps 50% of the time to A
or B another observation gives you a lot
of information uh information about the
world then that

policy can the policy that that leads
you to get that OB the informative
observation that has a high um epistemic
value

now that doesn't mean that policy
reaches into

a in the factorization of the base
graph right so it's it's not like every
single thing it's like saying well um
how could what the finger does influence
the

knee they're not connected directly it's
like well yeah but the whole generative
model is executed as a
whole so um

you can I mean to put it a different way
you can use expressions in the active
inference

ontology that associate different
variables that aren't necessarily
connected through an edge

here so these edges are not the only
possible sentences that you can
construct when talking about the
generative

model so you can talk about how G is
related to O right OB obviously because
observations are what are o Tilda are
what are being taken into account when
we're talking about G but there's no
line between G and

observations so this is just asking

um will those observations be
informative and and the observations
being
informative doesn't need to entail
update of the a
matrix I see I see yeah good point good
yeah or you know there there's two
people you can ask one of them he flips
a coin the other one right tells you the
truth so the epistemic value with going
to the person who gives you the good
information is
high even if you have a fixed
understanding of that
situation okay let's look at it in the
team A's
setting yeah yeah okay okay so so
technically okay technically the a
matrices are
tensors but it's a little bit of an
irregular
one okay so here we have these two kind
of parallel
A's on the the first and the second
[Music]
um uh context of a or first and the
second
factors mapping
um the actual
location here on the rows note that
there's this L and the
r this is because the hidden states are
in the cases of being on the right being
on the left and being in the starting
position those are all identity mappings
but now because we're in the black on
the right
setting the epistemic Q at the bottom

when you are physically in the bottom
you go to the hidden state which tells
you you're in the r context and there's
all zeros here that hidden State cannot
be reached but of course when the
context is Switched it's the exact
opposite there's a one here going to the
bottom Maps you to the it's the L State
and there is no way to reach the r
State and we also see a flipping of the
98s 98 is here on the bottom left and on
the right and here it's on the top left
on the right this is again just just to
show these matrices are necessary and
sufficient for specifying the
situation they describe literally
exactly what is happening all that your
generative model knows is what you
specify in these matrices there's no
other
information
so that's the a matrix this these this
is showing the two
contexts and it's
showing um the mappings
between um
location
and this kind of
like location
plus distinction between the messages
that the epistemic Q on the bottom could
provide and then also notice that um
when you're in the starting or the
bottom
position you're you're in the sort of
like dashed line like you're not finding
out anything but then when you're in but
then there's a 98% chance you have a 98%

sensory

accuracy there still is a 2%

mapping between um you know um if you

know if you're on the left in this

context and it's black there's there's

still a 2% chance it maps to

White

okay now we get to B so similarly we're

going to have two B

matrices um or two B tensors for each of

these State

factors B2 they're going to introduce

first because B2 the context stays

constant over time and can be

represented as an identity Matrix so

basically this is just keeping the same

thing through time within a

trial and this is where we get the the

true B

tensor

here this describes

location

changes as a function of movement

selection so we have from and to and

then there's four

[Music]

actions

um

B does all the

work let's look at the generative model

like observations are just data points

hidden states are hidden States um

they're they're just descriptors

proposed or actual Latent descriptors and

then this is just a sensory

mapping but all the work of the meaning

of embedded action is going to play out

through

b the dimensionality of
B is also higher because B is whatever
the dimensionality is of the state
Factor then as many slices as there are
policies yeah because each
policy is going
to change how the world changes now if
all of these slices were the
same that would be like saying whichever
policy you select the world is going to
change in the same way so in that
situation there's no real causal
efficacy or agency of that
agent yes so B is really where it's at
and in this example the B is just
given and fixed but there are other
examples that we could explore that you
can look up where there's B Matrix
learning so that's learning the
consequences of one's
actions through taking different
policies so that's a situation kind of
like a multi-armed bandit where you
actually have to learn the
mapping between different
things
um
preferences here C2 is the um the
context Associated preference
Vector so we have uh if there's new the
dashed line is neutral so this is these
three elements corresp
respond to these three elements 0 6
negative and
six and then these
five in the top C correspond to these
five um and then as they say there's
there's a slight they give a negative

one to the starting position just to
kind of bump it off and get it going
because if it really had no
preferences it it I
mean all of this comes into the
parameterization why is it six and not
nine and negative 9 why not nine and -2
that is the parameterization question so
when we're in a pedagogical setting and
we're just like looking to play around
that's when it's super helpful to have
the the code and like do parameter
sweeps or just by hand play around with
how different variables change the
outcome of the model um and then in
chapter nine it discusses more well how
would you go from a bunch of videos of
rats in the
Maze to inferring what the likeliest
parameters
are but here they're just
given um also just note that these
values
are given in
um a way that it's actually like e to
the
blank so the six
here has um actually a a
very
interpretable
meaning in terms of the ratio of the
preference so this is not just like six
arbitrary
points or like I don't know it is just
literally describing kind of like the
the odds
ratio
right

d in the
context um domain it's just one² one
one or it could have just said 0. 5.5
but they're they're
conveying that it has to sum to one
something must happen this is a
distribution and the values are the
same so uniform
distribution this is a perfect knowledge
over the starting
position right Daniel can ask a question
here
so this D_2 is half and half perity right
but this is not known to the agent who
makes the active inference right it is
known it it starts
out completely unsure it starts out with
a 50-50 belief but it's kind of funny
because it's like it's 100%
sure that it's
50/50
like same as a coin flip you're 100%
sure that it's
50/50 if you had a more sophisticated
model and and it is shown in a later
figure you could have a hyper
prior on D so you could ask how
confident am I about the prior you can
do that for any of
them okay okay so you have a hyper prior
that can uh characterize
 D_2 uh maybe character as some parameter
right so you can later on learn learn
those
parameters
exactly exactly any parameter that you
want to
learn in a bayian statistic setting you

need a hyper
prior if you don't if have if it's just
a fixed
parameter then it's like the equivalent
of the derac Delta
function it's just like a spike like you
have like it's just like a single
observation right yeah um and and it's
always a very interesting thing in this
case or in in your own generative models
like what is being made
explicit and then what is actually being
kept implicit for example like how does
the rat
know that it's going to get the
information at that
location right more more question how
does the r know it's half and a half
it's just given it's just given I mean
that's like asking how how does the rat
know that moving um to the left is going
to get it to the
left but but also um keep the uh figure
figure 9.2
in
mind weird it's like it
can't
9.1 we're making a map of the Rat here
is literally figure four figure 4.3 or
figure 7.3 right there D is not
shown so we're just describing its map
of its space we're not claiming to make
a total account of everything that the
rat
knows
um but yeah I mean we there's more to
say on this example then there's an i
example here's here in 710 here's the

prior on

D here's the lowercase letters or hyper

prior on

everything

then another example with a maze finding

agent and then a

hierarchical nested model where you you

put a

GM as a leaf in another

GM and that is shown in a nested example

so it's kind of a long chapter

actually but there because there's a lot

of examples that are kind of like

densely packed but just in our last like

minutes here what what does anyone find

interesting or what questions they still

have well um I still have one more

question in in the in the end of section

7.3

okay

um uh so it's uh two sentence above

7.4

okay yeah you're still at uh

okay okay um for I think further up um

so it talks about the neuron neuronal

activity uh okay yeah so the second to

the last sentence

uh it talk about field potentials and

neural

activities right uh in figure

7.2 yeah um so so my question is how

realistic is this um Chim

model uh comparing to the experimental

data so we have this is ass simulation

right so basically we constructed active

uh inference model using the data to

simulate the activities of the of of a

mouse in in the

that uh so my question is how does the simulation results compare to the uh actual biological experiment the activities of the mouse and how they um the brain the mouse brain um uh activities you know uh imply or or corresponds to this numerical simulation I'm not aware of them doing this in a rat doing the te- but it may exist we should look it up in other settings it Maps very very well now that's just like from from my perspective it's just a bonus because active inference is not developed as an account of brains the particular partition the basian mechanics all of the cognitive physics doesn't need to be about any specific system however as the empirical lines of work have shown various features of the model that are very non-trivial are recapitulated in Biometrics that suggests that the analytical solution and the biological embodiment have a convergence strategy an example of that is predictive coding predictive coding was invented by humans as a way to do video encoding because it turns out that frame differencing is the most efficient way to do video encoding like instead of just trying to compress every frame you basically compress the B Matrix because if something doesn't change you don't even need to mention it in the next slide so right predictive coding was a purely information theoretic consideration but it turned out to be

utilized in neural coding

situations

so I see it as positive where it aligns

but um where it doesn't align

then it it it it ALS it

doesn't it just means that there's

something else happening that that isn't

just narrowly

that yeah because you see in figure 7.2

they show the neural activities and the

firing Rays and local field potentials

so there must be some mechanism to

translate the outcome of the numerical

simulation into those neuro activities

right the the biological uh uh

embodiment of the numerical size um so

through what mechanism do they uh trans

transform this numerical into into the

uh into the field potentials for example

neuro

activities yeah it's all in the mat lab

code in the mat laab functions that are

used it describes it in more detail but

basically the lfps are associated with

with belief

updating how about fire race are they

associal with yeah decision you could

imagine different different firing rates

or different different I mean and this

has been studied extensively in the fmri

and EEG

setting um you know does does the um Erp

in the ERG or does the local field

potential associate with a more intense

stimuli with a more expected stimuli

with the less expected stimuli with

learning those are all hypotheses that

people have

explored and for different stimuli in
different

settings you see different patterns

[Music]

okay any last um comments

here um I mean I I don't want to keep us
too long but I just yeah I

still as the last bit I find the

Novelties

seeking uh rather interesting like just
the example it's roughly around page 145

146 like if you were to

model give it like if there were an

absence of preferences for an agent then

like I'm just quoting here the drive to

novelty resolution given by expected

free energy minimization leads our

simulated creature to avoid any

previously visited

locations um just like you know it

becomes like a purely like the model is

attempting to uh resolve expected free

energy if I understand this correctly by

just it's just seeking more and more

information it's going to places it

hasn't been before um there's nothing

pragmatic about returning to previous

places it's not attempting to realize

any preferences that say I need to go

back right so I just thought that's a

really nice kind of kind of emergent

property that we see whenever you leave

out preferences here and it it for me it

it kind of lays a lot of emphasis on

like the importance of of preferences

they're impact upon behavior in these

models and just maybe makes me think of

like how to you know maybe start with

preferences how to model those
appropriately because it can highly
impact um right like an agent's Behavior
yep great points and the strength of the
preferences

overall and and where that ends up on a
Continuum between novelty seeking and
basically ruthless pragmatic

indeed all right thank you all um we go
into chapter eight next time in the
later time slot so peace till next time
bye thank you Daniel Day